Syllabus — Class 4 Mathematics Olympiad

This maps the topic list for **SOF IMO Class 4** (primary target) against the two other most-common Class 4 maths olympiads in India — **Silverzone iOM** and **Unified Council UIMO** — plus a note on **Math Kangaroo** (international, different format). The core maths content is near-identical across all of them; what differs is how much weight each gives to pure logical reasoning vs. applied word problems.

1. SOF IMO — Class 4 (primary reference)

Section 1 — Logical Reasoning

- Patterns (number & figure series)
- Analogy and Classification (odd one out)
- Alphabet Test
- Coding–Decoding
- Ranking Test (ordering/comparison puzzles)
- Mirror Images
- Geometrical Shapes and Solids
- Embedded Figures
- Direction Sense Test
- Days and Dates, and Calendar
- Possible Combinations

Section 2 — Mathematical Reasoning

- Numerals and Number Names (up to 5–6 digit numbers)
- Number Sense: place value, expanded form, successor/predecessor, ascending/descending order, rounding off
- Roman Numerals
- Computation Operations: addition, subtraction, multiplication, division (including word-free direct computation)
- Factors and Multiples, simple patterns
- Fractions: like/unlike, comparing, equivalent fractions
- Measurement: Length, Weight, Capacity (and metric conversions)
- Time and Calendar

- Money
- Geometry: shapes, lines, angles (basic), symmetry
- Perimeter of regular and irregular shapes; introductory Area (counting unit squares)
- Data Handling: tally marks, pictographs, bar graphs, simple tables

Section 3 — Everyday Mathematics

- Same syllabus as Section 2, but every question is a real-life word problem (shopping, travel, sharing, measuring) rather than a direct computation — this is where arithmetic meets reading comprehension.

Section 4 — Achievers Section (HOTS)

- Higher Order Thinking Questions drawn from the Section 2 syllabus — usually 2–3 step problems (e.g. "find a number from a division clue, then use it in a second calculation").

2. Cross-olympiad topic map

Topic area	SOF IMO	Silverzone iOM	Unified Council UIMO
Number sense, place value, Roman numerals	✅ Sec. 2	✅ "Numbers and Their Operations"	✅ "Large numbers, Roman numerals"
Four operations (+ − × ÷)	✅ Sec. 2/3	✅	✅
Factors & multiples	– (light)	– (light)	✅ explicit unit
Fractions & decimals	✅ Sec. 2/3	✅ "Parts and Wholes"	✅
Length / weight / capacity / money	✅ Sec. 2/3	✅ "Money, Time and Calendar"	✅
Time & calendar	✅ Sec. 1 (dates) + Sec. 2/3	✅	✅
Geometry (shapes, symmetry)	✅ Sec. 1 & 2	✅ "Geometry"	✅
Perimeter & area	✅ Sec. 2/3	✅ "Area and Its Boundary"	✅ (within Maths)
Data handling	✅ Sec. 2/3	✅ "Graphical Representation of Data"	– (light)
Pure logical reasoning (coding, analogy, mirror images, direction sense)	✅ Sec. 1 (dedicated, 10 Q)	✅ "Reasoning and Aptitude"	✅ "Reasoning" (10 marks)

Topic area	SOF IMO	Silverzone iOM	Unified Council UIMO
Higher-order / critical thinking	✅ Sec. 4 "Achievers"	✅ "Scholar's Zone"	✅ "Critical Thinking"

Takeaway: practising the SOF IMO syllabus thoroughly (arithmetic + measurement + geometry + data handling + logical reasoning + HOTS) covers essentially all of Silverzone iOM and UIMO as well. This repo's assignments and sample papers are therefore built once, against the IMO list, and are valid prep for all three.

3. Math Kangaroo (optional, international)

Different format (24 MCQs, 1–5 points by difficulty, no separate reasoning section — reasoning is baked into the maths questions themselves) and no negative marking either, but a notably higher puzzle/spatial-reasoning flavour. Not required for Indian school olympiad prep, but a good stretch source once the core plan below is finished — see `assignments/12-math-kangaroo-stretch/` (15 questions, 3 difficulty tiers) and the "optional extension" note at the end of `03-practice-plan.md`.

Sources

- SOF International Mathematics Olympiad (IMO) Syllabus Class 4 — sofworld.org
- IMO Class 4 Syllabus for 2025-26 — Vedantu
- SilverZone IOM Syllabus — Vedantu
- IOM Syllabus for Classes 1 to 10 — Olympiad Success
- Unified International Mathematics Olympiad (UIMO) Syllabus — IndCareer
- Unified International Mathematics Olympiad — Vedantu
- Mathematical Kangaroo — Wikipedia